

Research on Multi-feature Extraction Method in EEG Signal of Motor Imagination

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Abstract—The feature extraction of EEG signals of motor imaging is a very important step in the brain-computer interface system. We usually use the event-related synchronization/desynchronization (ERS/ERD) phenomenon of EEG signals, and use common motor imagination EEG signal's feature extraction processing algorithms, including Common Spatial Pattern (CSP), Power Spectral Density (PSD), Autoregressive model (AR), Discrete Wavelet Transform (DWT) method and other methods to extract features of EEG signals. This paper expounds the basic principle of the algorithm, and finally uses the support vector machine (SVM) to classify the left-hand and right-hand motor imagery patterns after processing the four methods. Comparative experiments show that the feature extraction method of multi-feature fusion better represents EEG features, and a better classification effect can be achieved by using SUM classification. This study can meet the requirements of BCI for high recognition rate.

Keywords—*electroencephalogram(EEG); CSP; AR; PSD; SVM; Multi-feature Extraction*

I. INTRODUCTION

Brain-Computer Interface (BCI) is an interface that implements communication and control between the human brain and computers or other devices based on EEG signals. In recent years, due to the promotion of brain science, cognitive science, electronic measurement technology and computer information technology, the research of BCI has developed rapidly and attracted more and more attention.

Electroencephalography (EEG) is one of the most commonly used techniques for recording brain activity such as moving images. It has the characteristics of low signal-to-noise ratio and high classification error rate. Therefore, selecting the most discriminative feature is the key to improve the classification performance.

When the human body is performing motor imagination, the motor-sensing cortex on the opposite side of the brain is activated, the metabolism and blood flow in this area are increased, the activity of neurons is enhanced, the independence of EEG signals is enhanced, and the degree of synchronization is reduced, which in turn causes the energy of μ rhythm (8 ~ 13 Hz) and beta rhythm (14 ~ 30 Hz) is reduced; at the same time, the neuronal activity of the motor-

sensing cortex on the same side of the brain is suppressed, which in turn leads to an increase in the energy values of μ and β rhythms, as shown in Figures 1 and 2. It is called Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS). The discovery of ERD/ERS law provides an idea for the classification of motor imaging of EEG signals.

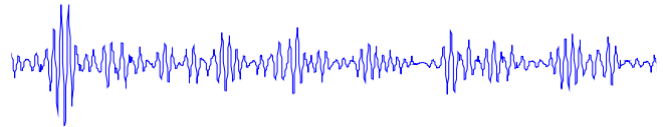


Figure 1. μ rhythm (8 ~ 13 Hz)

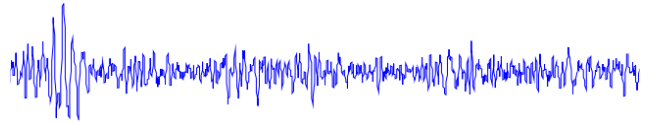


Figure 2. β rhythm (14 ~ 30 Hz)

EEG signal feature extraction and classification are one of the most challenging topics in BCI technology. Feature extraction, as a key step in EEG signal classification processing, plays a crucial role in the final classification result.

When the human brain imagines movement, ERP signals will appear in the left and right cerebral hemispheres, which will cause ERD/ERS phenomenon in the EEG power spectrum. For example, when the human brain performs a unilateral limb motor imaging task, the sensory cortex of the contralateral brain is activated, and the signal amplitude of the corresponding rhythm will decrease, while the ipsilateral cerebral cortex is in the blocked state, and the signal amplitude of the corresponding rhythm will be rise. Therefore, feature extraction pattern classification can be performed on the motor imaging EEG signals according to the ERS/ERD phenomenon of the brain. The distribution of ERS/ERD phenomenon in the cerebral cortex is shown in Figure 3.

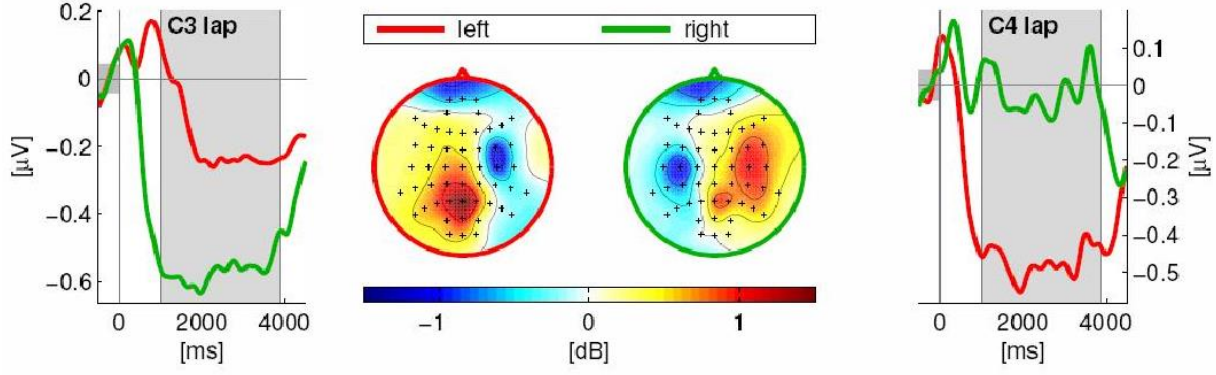


Figure 3. the distribution of ERD/ERS in the cerebral cortex when imagining movements of left and right hand

II. BASIC THEORY OF FEATURE EXTRACTION ALGORITHM

The feature extraction of EEG is the key link to realize BCI system. The following introduces several commonly used feature extraction algorithms.

A. CSP (Common Spatial Pattern)

The basic principle of CSP is to diagonalize the two covariance matrices and extract the spatial components of the two tasks to construct two types of spatial filters to maximize the difference between the two types. Its implementation steps are as follows.

(1) Calculate the covariance matrix for two types of signals.

$$C_i = \frac{E_i E_i^T}{\text{trace}(E_i E_i^T)} \quad (1)$$

Where: E_i represents the data matrix of left and right hands. $\text{trace}(E_i E_i^T)$ is a trace, indicating the sum of diagonal elements of the matrix.

(2) The average covariance matrix $\bar{C}$ of each class is calculated to form a composite covariance matrix of mixed space. The sum of the two types of average covariance matrix is as follows.

$$\bar{C} = \bar{C}_1 + \bar{C}_2 \quad (2)$$

(3) Perform eigenvalue decomposition on $\bar{C}$, that is $\bar{C} = U \lambda U^T$, where λ is the diagonal matrix of eigenvalues, U is the corresponding feature vector matrix.

(4) Perform whitening processing to construct a whitening matrix P .

$$P = \lambda^{-1/2} U^T \quad (3)$$

Thus, C_i is made into the following form.

$$S_1 = P \bar{C}_1 P^T, S_2 = P \bar{C}_2 P^T \quad (4)$$

(5) Since S_1 and S_2 have the same eigenvector, they are again decomposed by eigenvalues, and the formula will be obtained as follows.

$$s_1 = a \lambda_1 a^T, s_2 = a \lambda_2 a^T \quad (5)$$

Where a is a feature vector common to both, and $\lambda_1 + \lambda_2 = 1$.

(6) The feature vectors are sorted from large to small according to the size of the feature value. The m feature vectors arranged in the front are composed of a matrix B_1 , and the rest are B_2 , which form a pair of spatial filters.

$$W_1 = (B_1^T P)^T, W_2 = (B_2^T P)^T \quad (6)$$

E_i is filtered by using W_1 and W_2 , then E_i is changed to Y_i , thus $Y_i = W_i \times E_i$.

(7) Y_i is calculated as follows to form the eigenvalues.

$$f_p = \log \left[\frac{\text{var}(Y_p)}{\sum_{i=1}^{2m} \text{var}(Y_i)} \right] \quad (7)$$

Where $p=1,2,\dots,2m$. All f_p constitute a set of EEG eigenvalue.

B. DWT (Discrete Wavelet Transformation)

For any real number pair (a, b) , a is a non-zero real number, then the function of the following formula (8) is called a continuous wavelet function generated by the wavelet basis function $\psi(t)$.

$$\psi_{(a,b)} = \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-a}{b}\right) \quad (8)$$

This function depends on the scale parameter a and the time parameter b . If at the same time, the parameters a and b in formula (8) are discretized: $a_j = 2^j$; $b_k = k / 2^{-j}$, $j, k \in Z$, then the discrete wavelet is obtained:

$$\psi_{j,k}(t) = 2^{-j/2} \psi(2^{-j}t - k) \quad (9)$$

Where: j and k represent the frequency resolution and the amount of time shift, respectively. The discrete wavelet transform of function or signal $f(t)$ is defined as follows:

$$C_{j,k}(f, \psi_{j,k}) = 2^{-j/k} \sum_{n=-\infty}^{\infty} \overline{\psi}(2^{-j}t - k), \quad j, k \in Z \quad (10)$$

Among them, $\overline{\psi}$ is the conjugate operation of ψ . Then, the signal after wavelet transformation is reconstructed, that is, the inverter process can filter and compress the signal, which is defined as follows.

$$f(t) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} C_{j,k} \psi_{j,k}(t), \quad j, k \in Z \quad (11)$$

Using Mallat algorithm, the signal $f(t)$ is decomposed in finite layers.

$$f(t) = A_L + \sum_{j=1}^L D_j \quad (12)$$

Where L is the decomposition level, A_L is the low-pass approximate component, and D_j is the detail component at different scales. Suppose the sampling frequency of the signal $f(t)$ is f_s , then the sub-band corresponding to each component $A_L, D_L, D_{L-1}, \dots, D_1$ is:

$$\left[0, \frac{f_s}{2^{L+1}}\right], \left[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}\right], \dots, \left[\frac{f_s}{2^2}, \frac{f_s}{2}\right]$$

C. AR(Autoregressive model)

The AR model can be expressed by the following formula:

$$x(n) = -\sum_{i=1}^p a_p(i)x(n-i) + \varepsilon(n) \quad (13)$$

In the formula (13), $\varepsilon(n)$ is white noise with mean zero and variance σ^2 , p is the model order, and $a_p(i)$ is the autoregressive model parameter. Based on the final prediction error criterion, this paper determines the order as 5, and uses the Burg algorithm to extract its AR model coefficients and coefficient differences as features.

D. PSD (Power Spectral Densit)

The calculation steps are as follows:

(1) $x(n)$ is an infinite-length random sequence, and the interception length N becomes a finite-length sequence, which is called $x_N(n)$.

(2) Calculate the self-correlation function $\hat{R}_x(m)$ of $x_N(n)$ at $2m-1$.

$$\hat{R}_x(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_N(n)x_N(n+m) \quad (14)$$

Where, $m = -(M-1), \dots, -1, 0, 1, \dots, M-1$, and $M \leq N$.

(3) Find the Fourier transform of the correlation function to get the power spectrum, namely:

$$\hat{S}_x(e^{j\omega}) = \sum_{m=-(M-1)}^{M-1} \hat{R}_x(m) e^{-j\omega m} \quad (15)$$

III. MULTI-FEATURE EXTRACTION

A. Data set description

The EEG data set used in this study comes from the BCI Competition 2008 data sets 2b. The data set contains EEG signals from 9 different subjects. At the beginning of the experiment, the subjects were required to relax the whole body, and the linear distance between the eyes and the screen was 1 m. A corresponding direction arrow will appear as a prompt on the screen. The subject completes the corresponding hand movement imagination according to the prompt, and completes two sessions in two days, where each session contains 6 runs, each run contains 20 experiments, and 10 left hands, 10 times for the right hand, then a session contains a total of 60 times left-hand imaginations, 60 times for the right hand, a total of 120 times.

Each experiment lasts 8 ~ 9 s, the first 2 s is in a relaxed state; at the 2 s, there will be a prompt sound to indicate that the experiment is about to start, and at the same time, a cross cursor appears on the screen to enter the preparation state for 1 s; in the next 1.25 s, the direction arrow will continue to appear randomly on the screen; at the 4 s, the subject performs corresponding motor imagery according to the prompt for 3 s; on the 7 s, the subject stops imagining and enters a rest state. In this experiment, Ag/AgCl electrodes were used, and three electrodes were used to record the sampled EEG data, which were C3, Cz and C4 respectively, where C z was the reference electrode, and C3 and C4 electrodes included the richest information on imaging hand movements. In this paper, the data of C3 and C4 channels are used for analysis, the sampling frequency is 250 Hz, and the sampled data is preprocessed by a bandpass filter of 0.5 to 100 Hz.

B. Pattern recognition of EEG signals

In this experiment, SVM(support vector machine) are used for pattern classification. Support vector machine was first proposed by Vapnik. Its main idea is to establish a classification hyperplane as a decision surface, which maximizes the interval between positive and negative examples, and is often used in pattern classification problems.

The classification of support vector machine is achieved through the libsvm toolbox. In [0, 1], the mapminmax function is used to normalize the training samples and test samples, the K-CV method is used, and the SVMcgForClass function is used for rough optimization and fine optimization. First set the penalty parameters $c \min = -10$, $c \max = 4$; Set kernel function parameters $g \min = -4$, $g \max = 4$; Set cross experiment parameters $\nu = 3$; Set step parameters $cstep = 0.5$,

$gstep = 0.5$, $accstep = 0.9$, complete the fine optimization process and get the best cross-validation accuracy rate of 85%, Penalty parameters $c = 2.82843$, $g = 2.82843$. Finally, the best parameters and training samples are used to train the network, and the trained network is used to complete 120 experimental test sample classification predictions. By comparing the correct sample classification results, the correct classification of 100 experimental samples is obtained.

C. Analysis of experimental results

In this paper, the four feature extraction algorithms and support vector machines are used to classify the data, and the accuracy rates of single feature are obtained as shown in the figure 4 within 4~7.0 s.

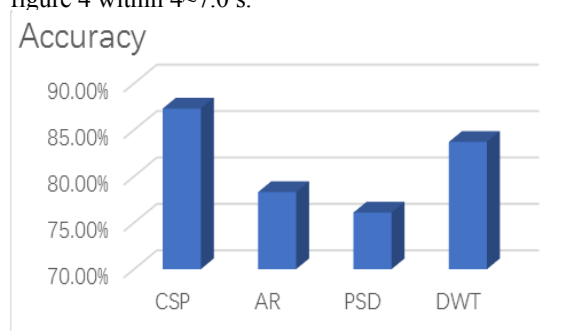


Figure 4. The accuracy rates of single-feature

It can be seen from Figure 5 that in terms of feature fusion, the accuracy of the three combinations is the highest, reaching 91.70%, and the accuracy of pairwise fusion is higher than that of their single features, which are all around 89%.

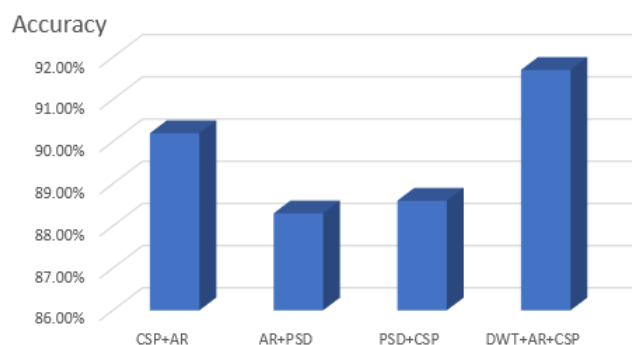


Figure 5. The accuracy rates of multi-feature

IV. CONCLUSIONS

This paper uses four algorithms, including Common Space Model (CSP), Discrete Wavelet Transform (DWT) Adaptive Regression (AR) model, and Power Spectral Density (PSD), to extract the features of the original EEG signal based on the ERD/ERS phenomenon. Support vector machine (SVM) is used to classify two types of motor

imagery EEG. The experimental results show that multi-feature extraction method is superior to single-feature extraction method. This paper provides a certain theoretical basis for BCI application feature extraction, and how to improve the accuracy on this basis is worth further research.

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